Systemadministration

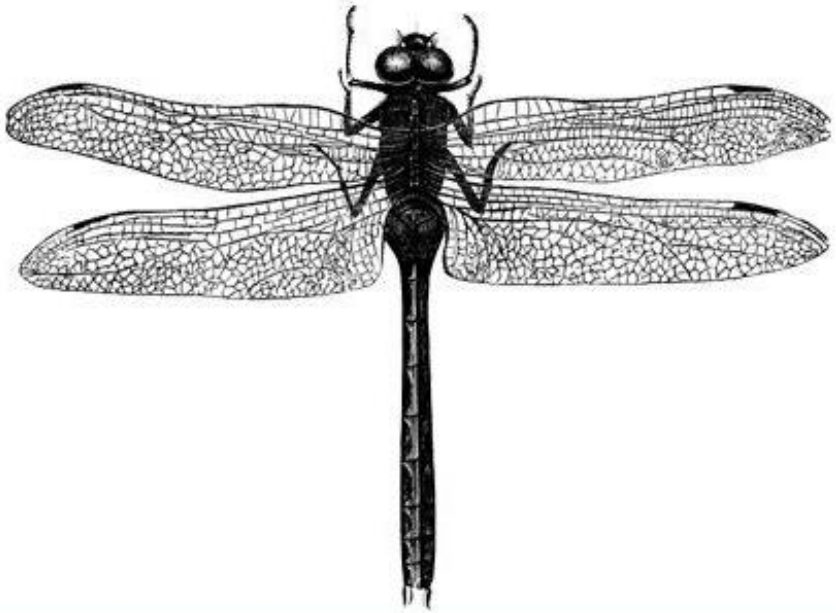
Linux

Local Names

Agenda

1. Hostname
2. Hosts
3. Domain Name System
4. Domain Names
5. DNS Components
6. DNS Resource Records
7. DNS Server with Forward & Reverse Zones

Editing /etc/hosts for a living



Everything is a Fscking DNS Problem

The Definitive Guide

O RLY?

Kris Buytaert

Hostname

hostname

Ett användarvänligt namn för en dator

```
nics@linux:~$ hostname
linux
```

Definieras i:

```
nics@linux:~$ cat /etc/hostname
linux
nics@linux:~$ sudoedit /etc/hostname
```

Appliceras vid uppstart

Hostname

Static hostname

- **/etc/hostname**

Transient hostname

- Dynamiskt, t.ex. ifrån DHCP
- Om det finns ersätter det **static hostname**

hostnamectl

systemd

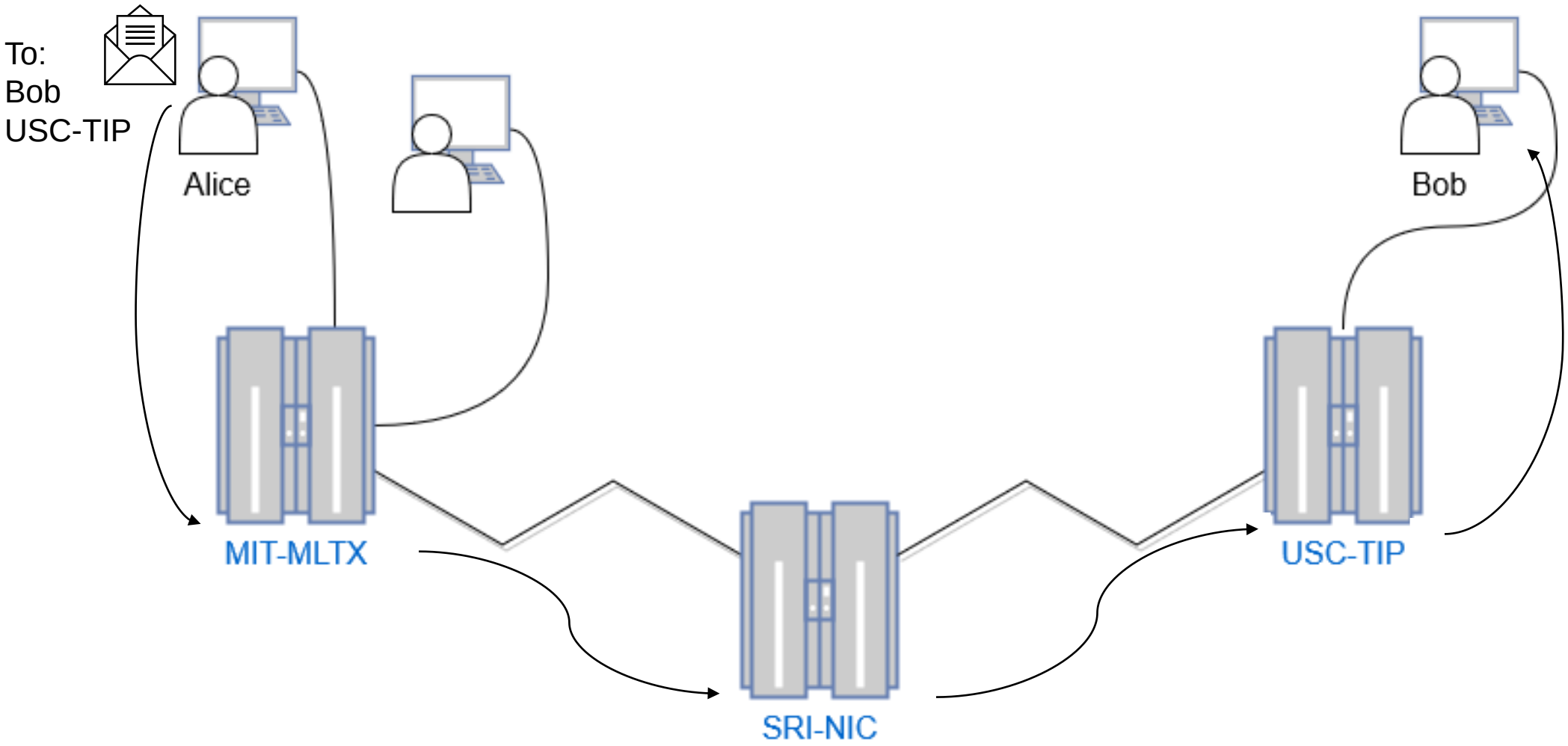
```
nics@demo:~$ hostname
demo
nics@demo:~$ cat /etc/hostname
demo
nics@demo:~$ sudo hostnamectl set-hostname linux
nics@demo:~$ cat /etc/hostname
linux
nics@demo:~$ hostnamectl
 Static hostname: linux
```

Ändrar både **static** & **transient hostname**…

Behöver ändå starta om!

Hosts

Networked Hostnames



THE HOSTS.TXT File

En lista på alla **networked hosts**

- hostname & tillhörande IP-adress

Formal Name	Nickname	Network Address
AMES-67		16
AMES-TIP		144
BBN-NCC	NCC	5
CMU-10		14
MIT-DMCG	DMCG	70
MIT-MLTX	MULTICS	6
MITRE-TIP		145

/etc/hosts

Idag:

```
nics@linux:~$ cat /etc/hosts
# IP address    canonical hostname        aliases
127.0.0.1       localhost.localdomain     localhost
::1             localhost.localdomain     localhost

192.0.2.10      linux.nicslab.se          linux
2001:db8:2::10  linux.nicslab.se          linux
```

/etc/hosts

Syntaxen är standardiserad

Distro-specifikt: Vilka namn som måste vara definierade

- se grundkonfiguration

Absolut minimum:

```
nics@linux:~$ cat /etc/hosts
# loopback IP --> localhost
127.0.0.1       localhost

# statisk IP --> hostname
192.0.2.10      linux
```

Ändra hostname: /etc/hostname & /etc/hosts

1. Ändra **hostname**
   /etc/hostname eller **hostnamectl**
2. Uppdatera **hosts**-filen med nytt **hostname**
   /etc/hosts
3. Starta om
   reboot

SRI-NIC HOSTS.TXT

"[…] it seems about time to put an end to the absurd situation where each site on the network must maintain a different, generally out-of-date, host list for the use of its own operating system or user programs."

RFC 606

Looking For A Solution

"What we hope for is an official list of host names"

RFC 289

Lösning: **DNS**

Datatracker

Host Name

RFCs (10 hits)

RFC 229 **Standard host names**	3 pages	1971-09
RFC 236 **Standard host names**	2 pages	1971-09
RFC 237 **NIC view of standard host names**	1 page	1971-10
RFC 247 **Proffered set of standard host names**	4 pages	1971-10
RFC 273 **More on standard host names**	3 pages	1971-10
RFC 280 **A Draft of Host Names**	3 pages	1971-11
RFC 289 **What we hope is an official list of host names**	3 pages	1971-12
RFC 606 **Host names on-line**	3 pages	1973-12
RFC 608 **Host names on-line**	4 pages	1974-01

Domain Name System

Domain **N**ame **S**ystem

Viktigt & komplext

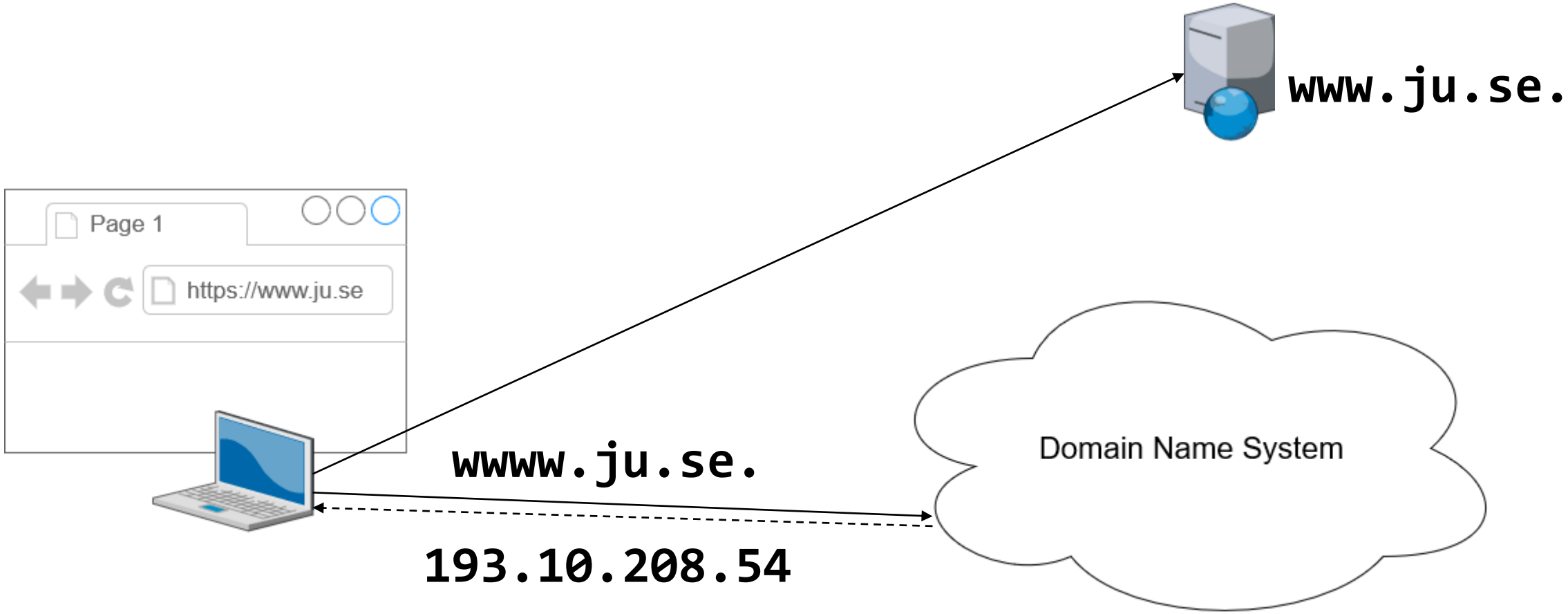
- Definierat i ca. 300 RFCer

UDP port **53**

- ’Vanliga’ **DNS**-frågor (<1,232 bytes)

TCP port **53**

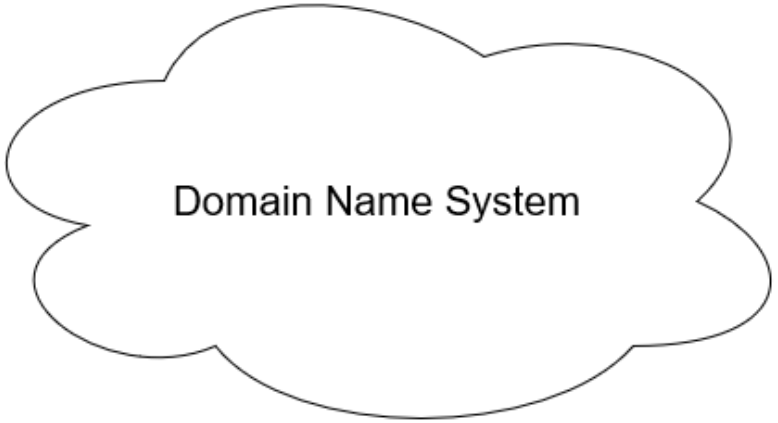
- ’Stora’ **DNS**-frågor (>1,232 bytes)
  - B.la. **Zone-transfers**, **TSIG**, **TXT**…


www.ju.se.
Page 1
https://www.ju.se
wwww.ju.se.
Domain Name System
193.10.208.54

Domain Names

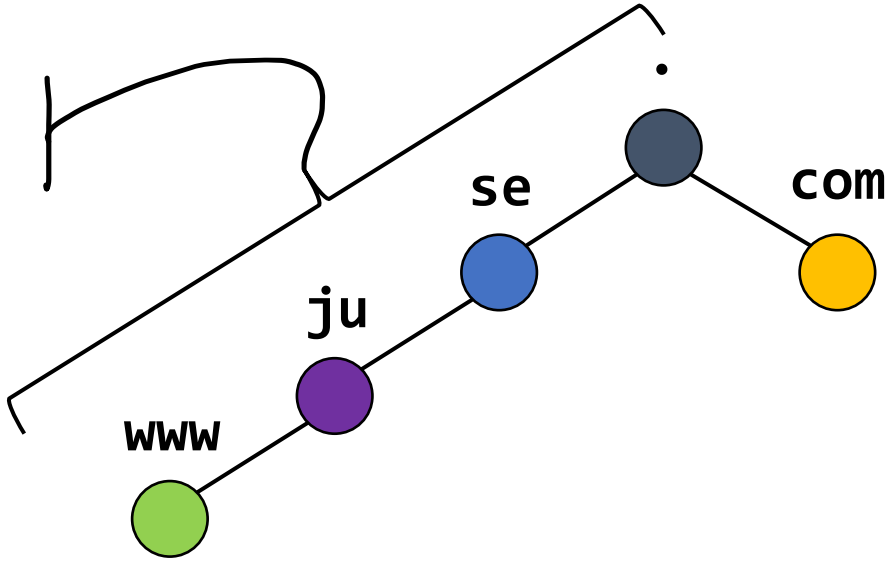
Domain Name Space

Namnrymd



En hierarkisk trädstruktur

Ex. **www.ju.se.**



Domain Names

Domännamn

Lite komplext & ofta missförstått…

Ex. **www.ju.se.**

- **F**ully **Q**ulified **D**omain **N**ame (**FQDN**):
  www.ju.se.
- **Domain Name**:
  ju.se.

En förenklad syn på domännamn

Ex. **www.ju.se**

- **Hostname**: www
- **Domain-name**: ju.se
- **FQDN (hostname + domain)**: www.ju.se

En mer korrekt syn på domännamn

- Each node in the **domain space** is a **domain**
- The **FQDN** of a **domain** is the sequence of nodes on the path to the **root** (.)

Ex.

www.ju.se.

3 **domains**

FQDN With or Without “.”?

“[…] a trailing period [is] to be used to denote the root, e.g., "a.b.c" and "a.b.c." are equivalent, but the latter is more explicit […]”

RFC 3969

www.ju.se = **www.ju.se.**

DNS-servrar är mer petiga:
FQDN måste avslutas med “.”

Local Domain Names

Domain	Description	Standard
local.	Reserverad av IANA för multicast DNS och ska därmed inte användas (ändå vanligast)	RFC 6762, Section 3 (2018)
home.arpa.	Reserverad av IANA för hemmabruk	RFC 8375 (2018)
intranet.	Förslag på interna domäner	RFC 6762, Appendix G (2018)
private.		
corp.		
home.		
lan.		
internal.	Preliminärt reserverad av IANA för interna domäner	RFC 6762, Appendix G & IANA

DNS Components

DNS Roles & Components

Name server

- Authoritative Name Server
- Caching Name Server
- Recursive Resolver
- Non-Recursive Resolver

DNS client

- Stub-Resolver

Authoritative Name Server

Auktoritativ namnserver

Ansvara för en **zon**

kan definiera:

- Domain names/domännamn
- Records/poster

inom sin **zon**

DNS Zone

En del utav **namnrymden**

Ex. nicslab.se & **dnlab.se**

- Början på 2 olika **zoner**
- **Auktoritativa namnservrar**: 193.10.31.164 & 193.10.31.190
  - Kan fritt definiera domännamn inom **zonen**: **bengt.dnlab.se**

Resolver

A type of **name server**

A **recursive resolver**

- *Resolves* domain names to e.g. IP addresses using the **DNS hierarchy**
- Caches DNS responses

A **non-recursive resolver**

- Only responds to requests for domains in a **zone** it is **authoritative** for
- Caches DNS responses

Stub-Resolver

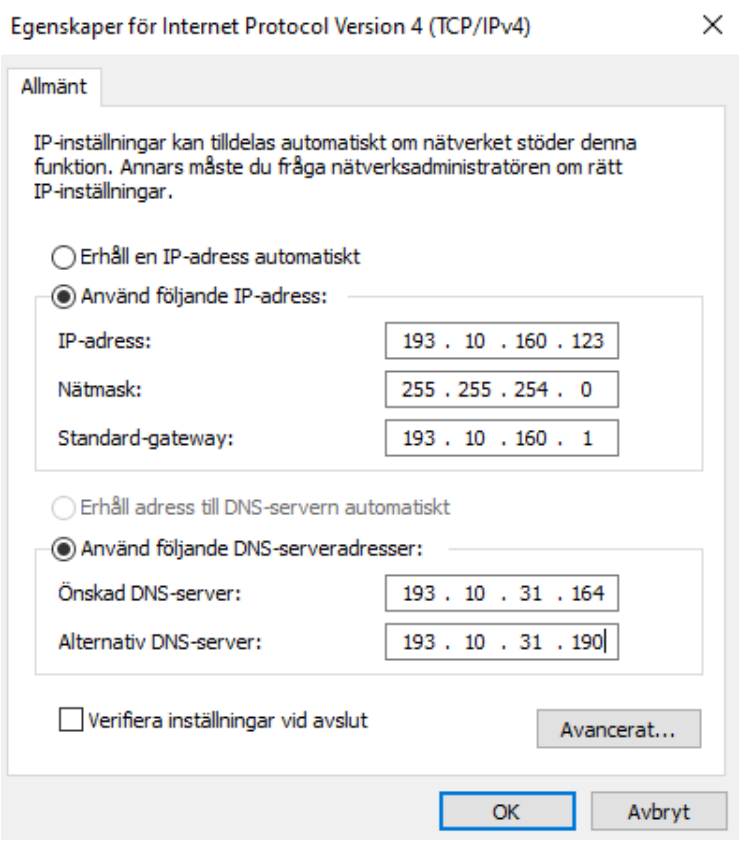
En klient som

- *Upplöser* domännamn till b.la. IP adresser genom att fråga en **rekursiv resolver**
- Caches DNS answeres

Stub-Resolver

Stub-Resolver

```
nics@linux:~$ cat /etc/resolv.conf
search nicslab.se
server 193.10.31.164
server 193.10.31.190
```



Linux Stub-Resolver Configuration

Nätverkskonfiguration hanteras på olika sätt beroende på **dist**

→ Se grundkonfiguration

system demon för att hantera DNS

systemd-resolved

[● ◀] systemd

```
nics@systemd:~$ resolvectl
Global
  resolv.conf mode: stub
Current DNS Server: 4.2.2.1
       DNS Servers: 4.2.2.1 4.2.2.2 208.67.220.220
```

DNS Resource Records

Resource Records

Poster

Information associerat med ett domännamn

Ex. IP-adressen till ett domännamn

```
????@ns1:/etc/bind/data/$ cat dnlab.se
<output omitted>
ns1.dnlab.se.        IN        NS   193.10.31.164
ns2.dnlab.se.        IN        NS   193.10.31.190
<output omitted>
```

Zone File

Zonfil

Poster definieras i en **zonfil**

Exakt en **zonfil** per **zon**

En **zonfil** måste:

- Inledas med en **SOA-post**
- Peka ut **auktoritativa namnservrar**

Common Resource Records

Vanliga poster

Record Type	Description
A	IPv4 address (domain name → IP)
AAAA	IPv6 address (domain name → IP)
CNAME	Domain alias (domain alias → domain name)
MX	Mail server for domain
NS	Authoritative name server for domain
SOA	Defines the beginning of a DNS zone
PTR	Reverse name (IP → domain name)

Resource Record Format

```
; Domain name        TTL        Class      Type       Value
```

Field	Description
Domain name	Domännamnet en post ska associeras med
TTL	**T**ime-**T**o-**L**ive, hur många sekunder en post ska cachas
Class	**IN**, INTERNET
Type	Resource record type
Value	Värdet som domännamnet associeras med

Type A Record

A-post

Mappar en IPv4-adress till ett domännamn

; Domain name	TLL	Class	Type	Value
web.nics.internal.	3600	IN	A	192.0.2.5

Type AAAA Record

AAAA-post

Mappar en IPv6-adress till ett domännamn

; Domain name	TTL	Class	Type	Value
web.nics.internal.	3600	IN	AAAA	2001:db8:2::5

Type CNAME Record

CNAME-post

Canonical **Name**

Fungerar som ett alias

; Domain name	TTL	Class	Type	Value
www.nics.internal.	3600	IN	CNAME	web.nics.internal.

Obs! alias placeras först

Type PTR Record

PTR-post

Pointer IP → domännamn

Ex. 192.0.2.5

```
; Domain name                TTL     Class   Type    Value
5.2.0.192.in-addr.arpa.     3600     IN      PTR     web.nics.internal.
```

Type NS Record

NS-post

Definierar **auktoritativ namnserver** för en **zon**

```
; Domain name        TTL     Class    Type       Value
nics.internal.       3600    IN       NS         ns1.nics.internal.
```

Utdrag från en exampel zonfile

```
; Domain name        TTL    Class   Type      Value
nics.internal.       3600   IN      NS        ns1.nics.internal.
nics.internal.       3600   IN      NS        ns2.nics.internal.

ns1.nics.internal.   3600   IN      A         192.0.2.11
ns2.nics.internal.   3600   IN      A         192.0.2.12

web.nics.internal.   3600   IN      A         192.0.2.5
web.nics.internal.   3600   IN      AAAA      2001:db8:2::5
www.nics.internal.   3600   IN      CNAME     web.nics.internal.
```

Utdrag från en exempel zonfil

Mer kortfattad men samma konfiguration

```
; @ = zone (nics.internal.)
$TTL 1H

@               IN      NS          ns1
@               IN      NS          ns2
ns1             IN      A           192.0.2.11
ns2             IN      A           192.0.2.12

web             IN      A           192.0.2.5
web             IN      AAAA        2001:db8:2::5
www             IN      CNAME       web
```

Type SOA Record

SOA-post

Start-**O**f-**A**uthority

Must be defined at the **zone apex**
= överst i **zonefilen**

```
domain   3600    IN     SOA      primary-name-server  responsible-person (
      serial      ; version number (YYYYMMDDNN)
      refresh     ; how often to perform zone transfers (seconds)
      retry       ; timer for retrying zone transfers (seconds)
      expire      ; timer to discard zone file (seconds)
      minimum     ; "negative response cache", NXDOMAIN TTL (seconds)
)
```

Example SOA Record

```
nics.internal. 3600    IN  SOA   ns1.nics.internal.   admin.nics.internal. (
      2024020503     ; serial  (YYYYMMDDNN)
      4H             ; refresh (14400)
      1H             ; retry   (3600)
      1W             ; expire  (604800)
      1H             ; minimum (3600)
)
```

…mer om **SOA-poster** senare, så länge:

Primary name server (MNAME)	FQDN för din primära namnserver	ns1.nics.internal.
Responsible Person (RNAME)	Oanvänt domännamn inom din zon	admin.nics.internal.
Serial	Dagensdatum + löpnummer	2024020503

DNS Server with Forward & Reverse Zones

Exempel konfiguration

Internal Name Server: Authoritative, Recursive & Caching

Konfigurationen delas ofta upp i flera filer

```
nics@ns1:~$ cat /etc/bind/named.conf
include "/etc/bind/named.conf.options";
include "/etc/bind/named.conf.local";
include "/etc/bind/named.conf.default-zones";
```

Officiella dokumentationen:
https://bind9.readthedocs.io/en/latest/index.html

Exempel Options

Internal Name Server: Authoritative, Recursive & Caching

```
nics@ns1:~$ cat /etc/bind/named.conf.options
options {
    pid-file "/var/run/named/named.pid";
    directory "/var/cache/bind";

    listen-on { any; };
    listen-on-v6 { any; };

    allow-query { any; };
    allow-query-cache { any; };
    allow-recursion {              // allow only local addresses
        127.0.0.0/24;
        ::1/128;
        192.0.2.0/24;
    };
    allow-transfer { none; };
    dnssec-validation auto;
};
```

Exempel Default-zones

Internal Name Server: Authoritative, Recursive & Caching

Editera inte:

```
nics@ns1:~$ head /etc/bind/named.conf.default-zones
// prime the server with knowledge of the root servers
<ouput omitted>

// be authoritative for the localhost forward and reverse zones,
//  and for broadcast zones as per RFC 1912
<ouput omitted>
```

Används bland annat för att upplösa publika domännamn

Exempel Local

Internal Name Server: Authoritative, Recursive & Caching

```
nics@ns1:~$ cat /etc/bind/named.conf.local
// block RFC 1918 reverse lookup
include "/etc/bind/zones.rfc1918";

// forward lookup zone
zone "nics.internal" IN {
      type primary;                // master in older versions
      file "nics.internal.zone";   // options: directory "/var/cache/bind";
      notify no;
};

// reverse lookup zone
zone "2.0.192.in-addr.arpa" IN {
      type primary;                // master in older versions
      file "192.0.2.zone";         // options: directory "/var/cache/bind";
      notify no;
};
```

Livedemo

Internal Name Server: Authoritative, Recursive & Caching

1. Installera **ISC BIND**, daemon: **named**
2. Starta & auto-start **named**
3. DNS-server 127.0.0.1
4. Konfigurera **BIND**
5. Kontrollera konfiguration
6. Applicera konfiguration
7. Kontrollera loggen
8. Justera nätverkskonfiguration

Livedemo

Internal Name Server: Authoritative, Recursive & Caching

```
nics@ns1:~$ # se status
nics@ns1:~$ sudo rndc status
nics@ns1:~$ # kontrollera konfiguration
nics@ns1:~$ sudo named-checkconf /etc/bind/named.conf
nics@ns1:~$ sudo named-checkzone nics.internal
 /var/cache/bind/nics.internal.zone
nics@ns1:~$ sudo named-checkzone 2.0.192.in-addr.arpa
 /var/cache/bind/192.0.2.zone
nics@ns1:~$ # applicera
nics@ns1:~$ sudo rndc reload
```

Livedemo

Internal Name Server: Authoritative, Recursive & Caching

Gör en förändring i zonfilen:

1. Ändra i zonfilen
2. Uppdatera **serial**!!!!
3. Kontrollera konfiguration
4. Applicera konfiguration

Till nästa gång

- Läs kurslitteraturen!
- Gör klart labb3!
- Halvtidsutvärdering
- Tisdag: demo

Tack för idag